

Subspace or not, and which clause decides

The **test** has three clauses — contains **0**, closed under addition, closed under scaling — and a candidate fails by failing one of them. Sorting by which clause is what turns a list of examples into a diagnostic.

PASSES – BY DEFINITION OR TRIVIALY

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1	The whole space the largest subspace	$V \subseteq V$ Closure holds because V is a vector space to begin with. Worth stating so that "subspace" is understood as at most V rather than strictly smaller — proper subspaces are the special case, not the definition.
2	The zero subspace $\dim = 0$; the smallest	$\{0\}$ All three clauses hold vacuously: $0 + 0 = 0$ and $c0 = 0$. Its dimension is zero, not one — the empty set is its basis , since 0 is dependent on its own and cannot belong to one.
3	Any span automatically a subspace	$\text{Span}\{\mathbf{v} , \dots , \mathbf{v} \}$ A combination of combinations is a combination, so closure is immediate. This is the cheapest way to prove something is a subspace — exhibit it as a span and the test is already passed. Lines and planes through the origin are the $k = 1$ and $k = 2$ cases.

PASSES – BECAUSE LINEARITY FORCES IT

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4	Null space subspace of $\mathbb{R}^n$	$\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$ If $A\mathbf{u} = \mathbf{0}$ and $A\mathbf{v} = \mathbf{0}$ then $A(c\mathbf{u} + d\mathbf{v}) = \mathbf{0}$ — one line, both closure clauses. The homogeneous right-hand side is what makes it work, which is exactly why $A\mathbf{x} = \mathbf{b}$ below fails.
5	Column and row spaces spans of the columns and rows	$\text{Col}(A), \text{Row}(A)$ Both are spans, so they inherit the result above without further argument. Together with the null space and left null space these are the four fundamental subspaces .

FAILS – AND WHICH CLAUSE

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6	Line or plane off the origin fails clause 1 – no 0	$\{\mathbf{x} + t\mathbf{v}\}, \mathbf{x} \notin \text{Span}\{\mathbf{v}\}$ Affine rather than linear. The origin test is the cheapest of the three and should be run first — it disqualifies this immediately, with no closure argument needed.
7	Solution set of $A\mathbf{x} = \mathbf{b}$ fails clause 1 – $A\mathbf{0} = \mathbf{0} \not\equiv \mathbf{b}$	$\mathbf{x} + \text{Null}(A), \mathbf{b} \not\equiv \mathbf{0}$ The null space translated off the origin. Not a subspace, but a coset of one — which is why the general solution has the shape it does and why its dimension still comes from the null space.
8	First quadrant of $\mathbb{R}^2$ fails clause 3 only	$\{(x,y) : x \geq 0, y \geq 0\}$ Contains the origin and is closed under addition — two clauses out of three. Multiplying by -1 leaves the set, and that alone disqualifies it. The instructive case: passing most of the test is not passing it.

Note the near-misses are near in different ways. An affine set is closed under nothing that matters because it misses the origin; the first quadrant contains the origin and survives addition but not negative scaling. Both are the same distance from being a subspace and fail for unrelated reasons — which is why all three clauses have to be checked rather than the one that looks likeliest.